

NEURAL BELLMAN OPERATORS

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This paper develops Neural Bellman Operators that construct continuations from their own fitted futures and return feasible economic policies. A constructive backend gives a primitive error account against the Bellman optimum and retains original action witnesses through exact neural compilation. For trained ReLU continuations, analytic innovation integration reduces scalar investment search to finitely many algebraic action witnesses. Directed whole-cell certificates protect a complete incumbent policy and propagate full-action errors through finite sweeps. A non-tensor residual cache and signed sensitivity of a smooth installed reference make verification executable without a full state-pair table or exponential shock-path enumeration. A prospective stopping theorem connects the returned policy to an expected-cost target, and a sample-split confidence box qualifies learned approximately null structure. In the original nonlinear investment family, a separately frozen sensitivity extension attains two- and ten-percent cost-reduction targets in all declared two-, four- and eight-dimensional services. All construction, verification and own inference are charged. Policy identities and comparative work separate the trained neural construction from benefits supplied by common verification; no uniform neural ranking is inferred.

KEYWORDS: Dynamic programming, Neural approximation, Policy certification, Information acquisition, Computational economics.

1. INTRODUCTION

A continuation approximation matters because it changes an economic decision. Its fitting error, the loss of its implemented policy, and the work needed to certify an improvement are different objects. This paper develops Neural Bellman Operators around that distinction. A construction uses its own fitted future values, returns

feasible action witnesses, and connects numerical resources to the cost of the policy actually implemented.

There are two linked numerical contracts in the same controlled economy. Under effective primitive moduli, the constructive backend returns a feasible policy with a prescribed error account against the original Bellman optimum. Theorems 2 and 3 retain the construction and its exact continuous original-owner implementation. A fitted backend instead proposes actions from learned features. Those proposals become deployable only after a whole-cell policy certificate is satisfied. A prospective service then stops when the complete policy attains a prespecified expected-cost target relative to an installed controller. This additional contract neither changes the Bellman optimum nor turns an empirical fitting error into an optimality guarantee.

The trained neural backend has an explicit computational structure. Under the model’s scalar uniform innovation, the expectation of a one-hidden-layer ReLU continuation is analytic. For the scalar investment action, the integrated objective is piecewise quartic, and a finite collection of rationally isolated cubic stationary points supplies exact action-lattice witnesses. This produces a terminating action search for a fitted nonlinear continuation without invoking another unconverged nonconvex optimizer. It is distinct from compiling a native min-plus envelope into an identical neural circuit. The verification theorem nevertheless applies to both neural and conventional candidates.

The policy account is implemented on a locally split, non-tensor observation partition. A one-state residual and value cache encloses the old policy’s continuation without a full table of state pairs. A signed sensitivity calculation for the smooth installed reference further retains economically useful continuation direction. We prove its validity and give explicit state-dimension, leaf-count and horizon costs. The original nonlinear investment family has a dimension-uniform smooth-reference stability bound, although the work needed for a chosen approximation accuracy can still grow with dimension. Every later discontinuous acquired actor is handled by the generic cache, not by differentiating away its jumps.

The economic experiment compares trained ReLU, quadratic and tree-based Bellman continuations in dimensions two, four and eight and horizons two, four and six. The original compiled-witness and tensor-FVI frontends remain in the two-

dimensional comparison. Each method pays for its own construction, verification, inference and durable output. Targets require two- or ten-percent expected-cost reductions from installed zero investment. A separately frozen signed-sensitivity extension attains both targets in every declared service; its preceding cache-only failures remain in the record. The complete-policy costs, nonterminal changes and all-in clocks determine what this means economically. They do not establish uniform neural superiority: common verification actions often determine the same returned policy for several generators, and quadratic FVI is an important low-work comparator.

A separate sample-split calculation learns approximately null directions under uncertain action exposure. The guarantee qualifies both the nuisance correction and the remaining Bellman comparison with the same validation confidence box. It therefore concerns inferred structure, rather than merely a hand-selected exact invariant, while retaining the maintained nonlinear drift and innovation model.

1.0.0.1. Related methods. Approximate policy iteration and fitted value iteration have established approximation-error analyses (Munos, 2003, Munos and Szepesvári, 2008, Scherrer et al., 2015). Conservative and safe improvement relate a policy update to an incumbent (Kakade and Langford, 2002, Pirodda et al., 2013, Metelli et al., 2021). Tree-based fitted iteration provides a strong conventional comparator (Ernst et al., 2005). We do not identify these general principles as neural innovations. Our contribution is the connection among constructive and trained own-future continuations, original feasible witnesses, acquired-state implementation, computable directed verification, and a prospective economic return with complete work accounting. The original action-contrast development also distinguishes residual transport from state bisimulation metrics, reward transformations and value-equivalent model classes (Ferns, Panangaden, and Precup, 2004, Ng, Harada, and Russell, 1999, Grimm, Barreto, Singh, and Silver, 2020).

The article develops this constructive theorem–implementation–economic chain for expected costs. The complete development article and proof companion retain the broader recursive-preference, temporal-self, diffusion and game applications under their original hypotheses. All earlier experiments, including adverse com-

parisons, are preserved there and in the source archive. Linear action-kernel and sampling identities in the present chain are not silently extended to those different economic evaluations.

2. THE ECONOMIC OPERATOR AND ITS POLICY ACCOUNT

Consider dates $t = 0, \dots, T$ and an invariant compact state domain $X \subset \mathbb{R}^d$. At date t , the action belongs to a nonempty compact set $A_t(x)$. The transition is $F_t(x, a, Z_{t+1})$, the current cost is $c_t(x, a)$, and the terminal cost is $g(x)$. Write $B_0 = 1$ and $B_t = \prod_{j < t} \beta_j$, with $\beta_j > 0$. An implemented policy includes its observation rule, action selection and feasible numerical realization, not just its stored parameters.

For a continuation v , define

$$Q_t(v; x, a) = c_t(x, a) + \beta_t \rho_t(v(F_t(x, a, Z))), \quad \mathcal{T}_t v(x) = \inf_{a \in A_t(x)} Q_t(v; x, a). \quad (1)$$

Throughout the deterministic comparison, the evaluations ρ_t are monotone and cash invariant on a common domain containing all functions and constant translates used below. They are therefore nonexpansive in the supremum norm. Continuous objectives, effective compact feasible correspondences and measurable minimizers, or measurable arbitrarily near-minimizers with their error charged, are assumed. The stipulated conditional recursion gives Bellman comparison against all admissible adapted controls. In the experiments ρ_t is conditional expectation, costs are bounded and innovations have their original continuous uniform law. The direct-cost inference in Section 8 is specific to expectation; it is not automatically an inference theorem for nonlinear recursive utility or games.

A Neural Bellman Operator is a construction and evaluation procedure for (1). It forms targets using its *own* fitted f_{t+1} , returns a neural continuation and a feasible policy, and records an economic error allowance. It does not receive the optimal continuation as a training label. Arbitrary trained networks can be evaluated by the first result below. The subsequent constructive backend specifies how a network and its actor are obtained without assuming successful nonconvex optimization.

2.1. Action-dependent errors and complete-policy loss

Let $H(a)$ be a true current cost and $\hat{H}(a)$ its continuation approximation. If $\hat{a}$ is δ -optimal for $\hat{H}$ on the same feasible action set, then

$$H(\hat{a}) - \inf_a H(a) \leq \delta + \text{osc}_a \{\hat{H}(a) - H(a)\}. \quad (2)$$

Indeed, compare $\hat{a}$ with an arbitrarily near-optimal true action and cancel the approximate objectives. The infimum limit gives the result. Additive continuation levels cancel; action-dependent discrepancies do not. Under a change in future policies, the continuation must be transported to the new economic problem before this inequality licenses reuse. The full development edition retains the original quantitative transport, recursive-domain and economic-application results. No transport bound is presumed valid after an unaccounted change of constraints or preferences.

PROPOSITION 1—One-sided policy sandwich: *Suppose, uniformly in state,*

$$f_t - \mathcal{T}_t f_{t+1} \leq u_t, \quad \mathcal{T}_t^\pi f_{t+1} - f_t \leq v_t, \quad \ell_T \leq f_T - g \leq u_T.$$

Then the actual feasible policy satisfies

$$0 \leq J_0^\pi - V_0^* \leq \sum_{t < T} B_t(u_t + v_t) + B_T(u_T - \ell_T). \quad (3)$$

The bound is invariant to date-specific constant shifts of the fitted continuations, provided the associated residual accounts are shifted consistently.

PROOF: Backward comparison and cash invariance give

$$V_t^* \geq f_t - \sum_{k=t}^{T-1} B_{t,k} u_k - B_{t,T} u_T, \quad J_t^\pi \leq f_t + \sum_{k=t}^{T-1} B_{t,k} v_k - B_{t,T} \ell_T,$$

where $B_{t,k} = \prod_{j=t}^{k-1} \beta_j$. The terminal claim is the stipulated enclosure; applying the Bellman or fixed-policy operator proves each inductive step. Subtraction gives (3). A shift $f_t \mapsto f_t + c_t^0$ adds $c_t^0 - \beta_t c_{t+1}^0$ to u_t and subtracts it from v_t . The terminal width is unchanged. Q.E.D.

A sharper valid residual account for the same continuation and the same deployed actor sharpens knowledge of the policy. It does not retrain that policy or change its realized cost. Conversely, subtracting two upper bounds in (3) does not identify the difference in actual costs. These distinctions govern both the historical comparisons and the new experiments.

3. CONSTRUCTING AND COMPILING A FEASIBLE NEURAL POLICY

Assume the primitive objective has, for feasible state-action pairs, state modulus $A_t^Q = C_{x,t} + \beta_t F_{x,t} L_{t+1}$ and action modulus $D_t = C_{a,t} + \beta_t F_{a,t} L_{t+1}$. Let

$$d_H(A_t(x), A_t(z)) \leq K_t^A \|x - z\|_1, \quad L_t = A_t^Q + D_t K_t^A, \quad L_T = L_g.$$

The Hausdorff condition and the primitive moduli imply that $\mathcal{T}_t f_{t+1}$ is L_t -Lipschitz whenever f_{t+1} is L_{t+1} -Lipschitz. All moduli and numerical integration procedures are supplied by the economic primitives, not fitted from successful outcomes.

At state nodes $x_{t,i}$ with covering radius h_t , form effective feasible k_t -nets. Evaluate their own-future objectives with certified error at most e_t . Minimize the computed labels and retain the corresponding *original* action $a_{t,i}$. A Borel repair $\Gamma_{t,i}(x)$ is feasible at x and satisfies

$$\|\Gamma_{t,i}(x) - a_{t,i}\|_1 \leq K_t^A \|x - x_{t,i}\|_1 + \zeta_t. \quad (4)$$

The allowance ζ_t concerns displacement; feasibility itself may not fail. Define

$$f_t(x) = \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}, \quad i_t(x) = \min \arg \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}. \quad (5)$$

Return $\pi_t(x) = \Gamma_{t,i_t(x)}(x)$. Terminal labels approximate g with error e_T and use the same cone construction with modulus L_g .

THEOREM 2—Own-future construction with feasible witnesses: *The functions in (5) have exact affine-ReLU realizations and Lipschitz moduli L_t independent of label error. The returned policy is feasible and measurable. It satisfies*

$$-e_t \leq f_t - \mathcal{T}_t f_{t+1} \leq D_t k_t + e_t + 2L_t h_t, \quad (6)$$

$$\mathcal{T}_t^\pi f_{t+1} - f_t \leq e_t + D_t \zeta_t,$$

and hence

$$J_0^\pi - V_0^* \leq \sum_{t < T} B_t(D_t k_t + 2e_t + 2L_t h_t + D_t \zeta_t) + B_T(2e_T + 2L_g h_T). \quad (7)$$

With effective covers, repairs and certified queries at the prescribed accuracy, allocating each nonnegative term a share of a positive economic tolerance gives a terminating finite construction. A fixed numerical precision or a capped finite experimental ladder need not contain that allocation.

PROOF: The selected node label lies between the node Bellman infimum minus e_t and that infimum plus $D_t k_t + e_t$. The Lipschitz property of the infimum implies that every cone lies above it minus e_t . At a nearest node its cone exceeds the infimum by at most $D_t k_t + e_t + 2L_t h_t$. At the attaining original index, the two feasible pairs and (4) give

$$Q_t(f_{t+1}; x, \Gamma_{t,i}(x)) \leq y_{t,i} + e_t + L_t \|x - x_{t,i}\|_1 + D_t \zeta_t.$$

This proves (6). The terminal residual belongs to $[-e_T, e_T + 2L_g h_T]$; Proposition 1 gives the loss bound. A finite minimum of equally Lipschitz cones is equally Lipschitz. The identities $|z| = \text{ReLU}(z) + \text{ReLU}(-z)$ and $\min(u, v) = u - \text{ReLU}(u - v)$ give the exact network. Finite lexicographic comparison and Borel repair give measurability. Effective refinement of the finitely many positively weighted terms yields the termination assertion. Q.E.D.

The Lipschitz-extension principle is classical (McShane, 1934). A native min-plus envelope and its exact ReLU circuit, using the same labels, witnesses and ties, are one method with two encodings. They return the same function and policy. Neither Theorem 2 nor a faster representation proves a uniquely neural capability or a training-success probability for Adam or L-BFGS. The mathematical increment is the complete feasible, own-future construction and implementation account; the empirical question is its work and policy value relative to conventional generators.

3.1. Original-owner compilation on a tensor cover

On a tensor cover $\mathcal{X} = \prod_{j=1}^d \{x_{j,0}, \dots, x_{j,n_j}\}$, put $S = \prod_j (n_j + 1)$. For arbitrary labels y_i , including inconsistent inexact labels, compute at each grid node v

$$(z_v, o_v) = \min_{\text{lex}, i} (y_i + L\|v - x_i\|_1, i).$$

The index o_v is the original label owner, not the transformed node v .

THEOREM 3—Continuous original-owner identity: *For a closed tensor cell C with corners $\mathcal{V}(C)$, and $x \in C$,*

$$f(x) = \min_{v \in \mathcal{V}(C)} (z_v + L\|x - v\|_1), \quad (8)$$

$$(f(x), i_*(x)) = \min_{\text{lex}, v \in \mathcal{V}(C)} (y_{o_v} + L\|x - x_{o_v}\|_1, o_v).$$

Coordinatewise forward and backward relaxations compute (z, o) in $O(dS)$ additions and comparisons and $O(S)$ storage. After locating the cell, a query uses $O(d2^d)$ arithmetic operations. Exact rational comparisons preserve all ties, including $L = 0$. Using the original owner’s repaired action preserves the uncompiled policy, its cost and its certificate exactly.

PROOF: The triangle inequality bounds $f(x)$ above by each corner expression. Choose the lexicographically first attaining site $i_*(x)$. In every coordinate choose the cell endpoint between x_j and $x_{i_*,j}$. At this corner distances add exactly. An owner of strictly smaller transformed value would improve the objective at x ; a tied smaller owner would contradict the original tie rule. Thus this corner has owner $i_*(x)$, proving both identities. Along a coordinate line, forward and backward relaxations respectively minimize over sites on either side, always comparing value–original-index pairs. Iterating over coordinates gives the separable distance transform and its stated work. Rational arithmetic and unchanged repair complete the claim. *Q.E.D.*

The nodal algorithm is the classical distance transform ([Felzenszwalb and Huttenlocher, 2012](#)); preserving the original action through continuous off-grid queries is the policy obligation. If there are $J_{t,i}$ node actions and M_t integration representatives, write $Q_t = M_t \sum_i J_{t,i}$. Literal dense cone evaluation costs $O(\sum_t dS_{t+1}Q_t)$ in

representation work. Compilation replaces this by $O(\sum_t [dS_{t+1} + Q_t d 2^d])$, plus cell location, primitive evaluation, integration, feasible repair, certificate and output. Rational bit lengths and state boxes crossing many cells incur additional costs. There is no dimension-free complexity assertion.

3.2. Acquired states and conventional actors

A b -bit coordinate sensor reports a dyadic cell and its midpoint $\hat{x}$. On the unit cube, $\|x - \hat{x}\|_1 \leq r(b) = d2^{-b-1}$. For the investment capacity $\kappa(x)$ below, clip the original action to the smallest capacity throughout that cell and round down to an action quantum Δ_t . This is feasible at every state in the cell. Relative to an exact true-state repair, its displacement is at most $2K_t^A r(b) + \Delta_t$. Transporting the witness cone from $\hat{x}$ to x adds at most $2L_t r(b)$, plus any certified selector objective error v_t . Thus the implemented-policy account adds

$$\sum_{t < T} B_t \{ (2L_t + 2D_t K_t^A) r(b) + D_t \Delta_t + v_t \}. \quad (9)$$

No continuity of the discrete owner selector is assumed. At uncertain numerical comparisons the simulator retains all possible actors, rather than choosing a favorable one.

For conventional FVI the continuation is multilinear on its realized tensor grid. Its actual modulus is bounded by the largest coordinate edge slope. A nearest-node actor uses a feasible node minimizer and true-state repair. Let h_t be the largest half-cell ℓ^1 radius, L_t^{gr} the Bellman graph modulus, and e_t the node query allowance. Its optimal residual upper bound is $D_t k_t + e_t + L_t^{\text{gr}} h_t$. Its selected-policy upper bound is $e_t + E_t$, where

$$E_t = \sup_x \{ y_{i(x)} + L_t^{\text{gr}} \|x - x_{i(x)}\|_1 - f_t(x) \}.$$

Partitioning cells by nearest-node boundaries makes the expression separately affine in each coordinate. Its maximum is therefore at a partition vertex; the implementation encloses all such vertices. Both uniform and curvature-refined FVI receive this sharper one-sided account, not an artificially doubled baseline. Their acquired-state allowance is derived separately in (33).

4. ALLOCATING RESOURCES TO AN ECONOMIC TOLERANCE

A single resolution parameter imposes an economic restriction that the Bellman problem does not impose. State coverage, action search and integration have different error coefficients and different effects on work. Refining all three at the same rate can therefore spend most of the service budget on the wrong source of error. We separate these choices while retaining the continuation, feasible-witness interpretation and all-state comparison developed above.

4.1. A separated primitive budget

Consider the investment family specified in Section 10.5, with $d \in \{2, 3, 4\}$ state coordinates, discount $\beta = 15/16$, and uniform coordinate grids with N intervals. At a state node use $K + 1$ feasible action fractions. Integrate the common scalar innovation using M equal-probability bins and their midpoints. The three integers need not agree. Let L_t be the primitive graph modulus of the constructed witness continuation, and let D_t be its action modulus. A downward numerical displacement of the action grid is denoted by δ_a .

The state and action covering radii satisfy $h_t = d/(2N)$ and $k_t \leq 1/(8K) + \delta_a$. The innovation lies in an interval of length $1/16$. Within a bin its expected absolute distance from the midpoint is $1/(64M)$. Since changing the scalar innovation by $z - z'$ changes the next state by at most $d|z - z'|$ in the ℓ^1 norm, a certified numerical query has an error allowance

$$e_t \leq \frac{\beta d L_{t+1}}{64M} + \zeta_t. \quad (10)$$

Here ζ_t covers arithmetic, interval evaluation and label rounding. It does not suppress the integration error. The terminal label allowance is e_T .

PROPOSITION 4—Separated-resource policy account: *Under the feasible-witness hypotheses, the returned policy satisfies*

$$0 \leq J^\pi - V^* \leq \frac{a}{N} + \frac{b}{K} + \frac{q}{M} + \eta, \quad (11)$$

where

$$\begin{aligned} a &= d \sum_{t < T} \beta^t L_t + d \beta^T L_g, & b &= \frac{1}{8} \sum_{t < T} \beta^t D_t, \\ q &= \frac{d}{32} \sum_{t < T} \beta^{t+1} L_{t+1}, & \eta &= \sum_{t < T} \beta^t (2\tilde{\xi}_t + D_t \delta_a) + 2\beta^T e_T. \end{aligned} \quad (12)$$

The primitive coefficients are independent of the computed labels. The leading number of innovation evaluations in target formation is

$$T(N+1)^d(K+1)M. \quad (13)$$

The compiled continuation additionally incurs its corner-evaluation and preprocessing costs. Pilot selection, policy extraction, verification, arithmetic bit lengths and output are separate charged components.

PROOF: Insert the state and action radii and (10) into the feasible-witness loss bound. The selected-policy upper account contributes the second copy of e_t , not a second action-search allowance. Collecting the coefficients of N^{-1}, K^{-1}, M^{-1} gives (12). There are $(N+1)^d$ state nodes, $K+1$ actions per node and M innovation representatives per action at each nonterminal date. No independence of fitting errors is used. *Q.E.D.*

The account explains why a common further state rung need not be a diagonal refinement. In the $T=3, p=4, d=2$ cell, the exact primitive coefficients are approximately $a=51.1569, b=2.3650$ and $q=1.0486$. The choice $(N, K, M) = (128, 16, 8)$ has account about $0.6786 + \eta$, below target one when the numerical remainder fits the remaining budget. The diagonal choice $(128, 128, 128)$ is more accurate, but its target-generation innovation count at that same state grid is $129 \cdot 128 / (17 \cdot 8)$ times larger. This is an arithmetic comparison of two declared allocations, not an observation of the unexecuted diagonal service or a runtime ratio. Both witness and conventional FVI receive the separated allocation in the experiment.

4.2. The optimal continuous allocation and its boundary cases

A useful planning problem minimizes the leading product $N^d KM$ subject to (11). This proxy isolates the primitive evaluation term; it is not the entire operation count or the measured wall clock. Write $E = \varepsilon - \eta > 0$ and suppose first that a, b, q are positive and all optimal resources exceed their required lower bounds.

PROPOSITION 5—Error shares and clipped allocation: *The unique interior minimizer of the continuous proxy has*

$$\frac{a}{N} = \frac{d}{d+2}E, \quad \frac{b}{K} = \frac{1}{d+2}E, \quad \frac{q}{M} = \frac{1}{d+2}E. \quad (14)$$

For lower resource bounds $r_i \geq \underline{r}_i > 0$, put $c = (a, b, q)$, $p = (d, 1, 1)$ and $\bar{x}_i = c_i / \underline{r}_i$. If $E < \sum_i \bar{x}_i$, there is a unique positive λ such that

$$x_i = \min\{\bar{x}_i, p_i \lambda\}, \quad \sum_i x_i = E; \quad r_i = c_i / x_i. \quad (15)$$

These resources minimize the constrained continuous proxy. If $E \geq \sum_i \bar{x}_i$, all lower resource bounds suffice. Coordinates with zero error coefficient remain at their resource lower bound. When dyadic lower bounds are used, rounding each optimal resource upward to a power of two preserves the error guarantee and multiplies this proxy by less than 2^{d+2} .

PROOF: Set $x_i = c_i / r_i$. Minimizing the resource product is equivalent to maximizing $\sum_i p_i \log x_i$ over $0 < x_i \leq \bar{x}_i$ and $\sum_i x_i \leq E$. Strict concavity and the multiplier conditions give p_i / x_i equal on unconstrained coordinates and larger on capped coordinates. This is precisely (15). The sum of its right sides is strictly increasing until all caps bind. The unconstrained case yields (14). Upward dyadic rounding never enlarges an error term and multiplies $N^d KM$ by less than the stated factor. Neither argument bounds rational-arithmetic bit cost or computer time. *Q.E.D.*

The optimization is a standard separable resource-allocation calculation. Its role here is to expose the error coefficients of the actual own-future policy construction, including integration and implementation, rather than to claim novelty for the multiplier argument. A finite experimental menu need not contain the continuous optimum. The prototype uses a common, frozen dyadic menu for all generators

and records all of it. It does not describe this menu as a globally optimal tuning policy for either FVI or the witness method.

4.3. A complete tolerance frontier

For method m , let a prospectively fixed ladder return certificates $g_{m,r}$ and cumulative costs $w_{m,r}$, including failed earlier rungs. Define

$$r_m(\varepsilon) = \min\{r : g_{m,r} \leq \varepsilon\}, \quad W_m(\varepsilon) = w_{m,r_m(\varepsilon)}. \quad (16)$$

An empty set is a capped failure, not an interpolated success. Sort the finite union of all certificate values and zero. On each left-closed, right-open interval between successive values, every method's first crossing and complete work are constant. Evaluating the first-crossing rule at the left endpoint therefore reconstructs the entire positive-tolerance frontier exactly, even when successive certificates are not monotone. An unbounded final interval is retained. Repetitions have separate costs, while their mathematical checkpoints must be checked for identity.

This finite partition supplies the appropriate interpretation of a threshold advantage. At a fixed resolution the witness method may cost more. At a tolerance where it crosses one rung earlier, the smaller certificate may more than offset that cost. Whether this happens, and over which tolerance interval, is determined by (16); it is not determined by a favorable single threshold. Under an additional homogeneous model $g_m(n) = A_m/n$ and $w_m(n) = C_m n^\kappa$, the continuous frontier ratio is $(C_W/C_F)(A_W/A_F)^\kappa$. This last expression is a conditional explanation, not a fitted scaling law for the experiments. The release supplies the exact finite partition, sensitivity around one, complete prefixes and separate fixed-resolution observations.

5. THE NONLINEAR INVESTMENT FAMILY

The economic law is the original dimension-normalized investment family. For $x \in [0, 1]^d$, scalar action $0 \leq a \leq \kappa(x)$, and cyclic coordinate $v = j + 1$,

$$\kappa(x) = \frac{1}{8} + \frac{1}{8d} \sum_j x_j, \quad (17)$$

$$F_j(x, a, Z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_v}{8} + \frac{x_j(1 - x_v)}{16} + \gamma_j a + s_j Z,$$

where (γ_j, s_j) alternates between $(1/2, 1)$ and $(1/4, -1)$ and Z is uniform on $[-1/32, 1/32]$. The initial state is uniform on the cube and $\beta = 15/16$. The stage and terminal costs are

$$c(x, a) = \frac{2}{d} \sum_j (x_j - 5/8)^2 + \frac{1}{4d} \sum_j (x_j - x_{j+1})^2 + 2 \left(\frac{1}{2} - \frac{2}{d} \sum_j x_j \right)_+^2 + a^2 + 4a^4, \quad (18)$$

$$g(x) = \frac{4}{d} \sum_j (x_j - 5/8)^2 + \frac{1}{4d} \sum_j (x_j - x_{j+1})^2 + 2 \left(\frac{1}{2} - \frac{2}{d} \sum_j x_j \right)_+^2.$$

The original coefficients and continuous laws are unchanged. The domain is invariant: the drift lies in $[1/16, 11/16]$, $a \leq 1/4$, $\gamma_j \leq 1/2$, and the innovation magnitude is at most $1/32$. The results are for this theoretical economy, not calibrated installation fees or estimated welfare magnitudes.

6. TRAINED NEURAL CONTINUATIONS AND EXACT INNOVATION INTEGRATION

The constructive backend supplies a feasible reference without relying on the success of nonconvex optimization. A trained backend addresses a different computational question: whether features fitted to the operator's own continuation targets produce useful candidate actions at a prescribed resource budget. These two backends are not identified with one another. In particular, fitting the hidden weights below is not compilation of the min-plus construction in Theorem 2.

Let the transition have the one-factor form

$$F_t(x, a, Z) = \mu_t(x, a) + s_t(x, a)Z, \quad Z \sim \text{Unif}[-r, r].$$

The deterministic map μ_t may be nonlinear and the state may be multidimensional. Consider a fitted continuation with m hidden units,

$$h(y) = c + v^\top y + \sum_{j=1}^m u_j (w_j^\top y + b_j)_+. \quad (19)$$

Both hidden and output coefficients may be data dependent. Once a candidate has been constructed, its stored coefficients specify an exact numerical function;

uncertainty about whether training found a good function is not an arithmetic uncertainty about the function that was stored.

LEMMA 6—Integrated neural continuation: Put $z_j = w_j^\top \mu_t(x, a) + b_j$ and $R_j = r|w_j^\top s_t(x, a)|$. Then

$$P_t^a h(x) = c + v^\top \mu_t(x, a) + \sum_{j=1}^m u_j \Phi(z_j, R_j), \quad (20)$$

where $\Phi(z, 0) = z_+$ and, for $R > 0$,

$$\Phi(z, R) = \begin{cases} 0, & z \leq -R, \\ (z + R)^2 / (4R), & -R < z < R, \\ z, & z \geq R. \end{cases}$$

For fixed stored coefficients and evaluable primitives, a point expectation uses $O(md)$ arithmetic operations in addition to evaluating μ_t and s_t . Outward evaluation on a state-action box gives a valid enclosure. There is no innovation-grid discretization remainder in this analytic expectation.

PROOF: The scalar random variable $w_j^\top s_t Z$ is uniform on $[-R_j, R_j]$, or is identically zero. Integrating its positive part gives the displayed three cases. Linearity of expectation gives (20); the affine innovation term has mean zero. Each feature requires two inner products and a bounded number of scalar operations. The function Φ is nondecreasing in each argument on $R \geq 0$: in the interior its radius derivative is $(R^2 - z^2)/(4R^2) \geq 0$, and outside the interior it is constant in the radius. Endpoint evaluation with directed arithmetic, enclosing an uncertain branch rather than choosing a favorable branch, therefore preserves containment. Negative output weights must reverse interval endpoints in the usual way. Q.E.D.

The formula is a property of this neural class under the stated innovation law, not a general exact-integration result for arbitrary deep networks or unknown kernels. Quadratic critics also have analytic expectations here; tree critics use valid bin enclosures. Thus eliminating quadrature for the neural backend is a specific computational facility, not a claim that conventional approximation necessarily needs

quadrature. On the active hinge region the equivalent expression $(z + R)^2 / (4R)$ avoids subtracting two nearly equal squares. At $R = 0$ the separate branch is essential.

The fitted construction proceeds backward. At each date it draws training states, forms a finite feasible action menu, evaluates current cost plus the integrated *own* subsequent critic, minimizes these labels, and fits the next critic. Its training labels are not evaluations of the optimal Bellman continuation. Fitted value iteration with general function classes has a substantial literature; see [Munos and Szepesvári \(2008\)](#) and the tree-based construction of [Ernst et al. \(2005\)](#). Our use of training is accompanied by a separate whole-cell verification obligation. No training mean-square error is substituted for that obligation.

In the executed neural construction the hidden layer has width 32. All hidden and output coefficients receive 120 full-batch Adam updates, followed by a ridge fit of the affine and output coefficients on the learned features. Coefficients are rounded to 24 binary fractional bits. The verifier evaluates this quantized function, not a hypothetical unrounded optimizer. The fitted hidden weights, targets, states, seeds and quantized output are retained. A fitted nonlinear network, a quadratic continuation, an ExtraTrees continuation, and the two original constructive frontends are therefore identifiable, distinct candidate generators. Whether their additional proposals change the deployed policy is an empirical question that we report separately from their fitting errors.

6.1. *Certified action search for the fitted neural continuation*

The same integrated representation also permits a constructive action search, rather than a second unconverged optimization problem. Fix a reported state in the original economy, so the deterministic next state is $\mu(x, 0) + \gamma a$. Let the feasible deployed actions be $\{0, \Delta, \dots, M\Delta\}$, with $M\Delta$ no greater than the whole-cell lower capacity. Interpret stored coefficients and the reported state as exact dyadic numbers.

THEOREM 7—Algebraic neural action witnesses: *For a continuation of the form (19), its integrated Bellman objective in the original scalar investment action has at most*

2m interior breakpoints. Between consecutive breakpoints it is a quartic polynomial

$$4a^4 + q_2a^2 + q_1a + q_0,$$

with rational coefficients. An exact minimizer on the deployed action lattice can be found among at most $22m + 13$ candidate indices, independently of the number of lattice points. Construction requires $O(md + m^2 + m \log m)$ rational arithmetic and comparison operations, plus $O(m)$ cubic-root isolation problems at the action precision. Coefficient bit lengths and root-isolation work must also be charged. The selected action is an original fitted-critic action witness, not a min-plus encoding.

PROOF: For feature j , write its preactivation mean as $z_j + s_j a$ and its innovation radius as R_j . When $s_j \neq 0$, branch changes occur only at $(\pm R_j - z_j)/s_j$; for $R_j = 0$ these coincide. Retain those breakpoints inside the feasible interval and its two endpoints. Lemma 6 makes each feature quadratic, affine or zero between breakpoints. Adding $a^2 + 4a^4$ and the affine part of the critic gives the displayed polynomial. Its derivative is $16a^3 + 2q_2a + q_1$ and has at most three real roots in each of at most $2m + 1$ pieces.

Between stationary points and breakpoints the objective is monotone. Hence a lattice minimizer lies immediately to either side of a breakpoint or a stationary point, or at a feasible endpoint. Isolate each stationary root in a rational interval of width at most $\Delta/4$ and include every index from the floor of its lower endpoint divided by Δ through the ceiling of its upper endpoint divided by Δ . This contributes at most three indices per isolated root and contains the required neighboring lattice points. The breakpoints and endpoints contribute at most $2(2m + 2)$ indices. Thus $9(2m + 1) + 4m + 4 = 22m + 13$ is a valid bound. Extra indices and duplicate candidates do not invalidate minimization. Exact rational evaluation and lexicographic value-action comparison settle ties.

Computing feature coefficients costs $O(md)$, sorting the breakpoints costs $O(m \log m)$ comparisons, and literal formation of the polynomial on each piece costs $O(m^2)$ operations. Rational isolation of the finitely many cubic roots terminates at positive action precision, including multiple roots. This proves the construction and

the stated separation between arithmetic counts and bit-dependent root-isolation work. Q.E.D.

The solver is implemented with rational root isolation and is checked against exhaustive exact lattice minimization, including saved fitted critics from each of the three state dimensions. It minimizes the stored critic's objective at the reported state. Whole-cell safety still requires the true-advantage certificate; critic optimality alone does not justify deployment. The recorded economic experiment used its prospectively fixed 17-action construction menu. Its policies and measured clocks are not retroactively replaced by this new action solver, and its cost intervals are not attributed to untested replacement policies.

7. A COMPLETE-POLICY CERTIFICATE ON NON-TENSOR ACQUIRED STATES

For the remainder of the new verification and stopping results, the recursion is the bounded finite-horizon expected-cost model. Let $\mathcal{C}$ be a finite measurable partition, with closed boxes used only to verify the cells. Boundary ties follow a specified observation rule. A deployed action is constant on each reported cell and feasible at every state in its closure. A verification box meeting several cells must retain every possible future action. The argument does not require a tensor partition or a continuous actor.

Fix a complete incumbent π^k and define its true advantage

$$A_t^k(x, a) = c_t(x, a) + \beta_t P_t^a J_{t+1}^{\pi^k}(x) - J_t^{\pi^k}(x).$$

The incumbent has advantage zero. For each cell C , let a_i be robustly feasible candidates and let $\mathcal{B}_t(C)$ be a finite interval cover containing every action feasible at every $x \in C$. The cover may include additional actions; its bounds must remain valid on the feasible subset. Suppose verified endpoints satisfy

$$A_t^k(x, a_i) \leq u_i(C), \quad \ell_B(C) \leq A_t^k(x, a) \quad (a \in B \cap A_t(x)).$$

Set $U_t^k(C) = \min\{0, \min_i u_i(C)\}$ and $L_t^k(C) = \min\{0, \min_B \ell_B(C)\}$. Select the candidate attaining U , retaining the incumbent on a tie. All dates are computed against the same old policy and adopted simultaneously.

THEOREM 8—Directed policy improvement: *Under these conditions the new policy is feasible and*

$$J_t^{\pi^{k+1}}(x) \leq J_t^{\pi^k}(x) \quad (21)$$

for every state and date. With $E_t^k = \sup_x (J_t^{\pi^k}(x) - J_t^*(x))$ and $\epsilon_t^k = \max_C (U_t^k(C) - L_t^k(C))$,

$$E_t^{k+1} \leq \beta_t E_{t+1}^k + \epsilon_t^k. \quad (22)$$

For $B_{t,j} = \prod_{i=0}^{j-1} \beta_{t+i}$, $B_{t,0} = 1$, and $m = \min\{K, T - t\}$,

$$E_t^K \leq \sum_{j=0}^{m-1} B_{t,j} \epsilon_{t+j}^{K-1-j} + B_{t,m} E_{t+m}^{K-m}.$$

The final term vanishes when $K \geq T - t$. Exact full-action greedy certificates thus recover the original Bellman optimum within T passes. The coefficient one multiplying the directed error cannot be reduced uniformly.

PROOF: The selected advantage is at most $U \leq 0$. Starting with the common terminal value, backward monotonicity proves (21). The full-action lower cover gives

$$A_t^k(x, \pi_t^{k+1}(x)) - \inf_{a \in A_t(x)} A_t^k(x, a) \leq U_t^k(C) - L_t^k(C).$$

Replacing the new continuation by the old one using safety, and comparing with an arbitrarily near-optimal Bellman action, yields (22). Taking the infimum limit avoids an attainment assumption. Iterating the inequality gives the displayed finite-horizon expansion. For sharpness take one decision date, an incumbent costing $\delta > 0$, and a feasible unproposed action costing zero. An exact incumbent upper endpoint is zero and an exact covering lower endpoint is $-\delta$. Retaining the incumbent gives both error and directed allowance equal to δ . Q.E.D.

Theorem 8 restates the original directed result in the partition notation used by the new implementation. It is deliberately representation neutral. Its premise is stronger than checking a finite candidate menu: without the full-action cover, a safe improvement need not have a small gap from the continuous-action optimum. A

change of acquisition partition between passes is permitted when the old policy is transferred exactly to its child cells; the complete incumbent remains the same function.

7.1. A one-state residual and value cache

Let $h_T = g$ and h_t be any stored critics, including fitted networks. Write $e_t = J_t^\pi - h_t$ and

$$r_t = c_t(\cdot, \pi_t) + \beta_t P_t^{\pi_t} h_{t+1} - h_t, \quad e_t = r_t + \beta_t P_t^{\pi_t} e_{t+1}.$$

A cellwise residual interval $\mathcal{R}_t(C)$ encloses r_t throughout the cell. If $\mathcal{E}_{t+1}(C')$ encloses the next error, each innovation bin maps a current box and action into a next-state box. The hull of $\mathcal{E}_{t+1}(C')$ over all cells intersecting that box encloses every possible next error. Denote the probability-weighted sum of those hulls by $\mathfrak{P}_t^a \mathcal{E}_{t+1}(C)$. Bin probabilities refer to the original continuous law, not an empirical law of midpoints.

PROPOSITION 9—Cached residual verification: *Starting from $\mathcal{E}_T = \{0\}$, the backward recursion*

$$\mathcal{E}_t(C) = \mathcal{R}_t(C) + \beta_t \mathfrak{P}_t^{\pi_t} \mathcal{E}_{t+1}(C) \tag{23}$$

encloses e_t on every cell. An independently computed direct value interval $\mathcal{V}_t(C)$ may be intersected with $h_t(C) + \mathcal{E}_t(C)$, and $\mathcal{E}_t(C)$ with $\mathcal{V}_t(C) - h_t(C)$. These intersections preserve containment. For feasible a, b the sum

$$c_t(C, a) - c_t(C, b) + \beta_t \{ (P_t^a - P_t^b) h_{t+1}(C) + \mathfrak{P}_t^a \mathcal{E}_{t+1}(C) - \mathfrak{P}_t^b \mathcal{E}_{t+1}(C) \}$$

encloses the fixed-incumbent action difference. Intersecting it with the analogous direct-value difference gives a valid directed interval. Identical actions have the exact interval $\{0\}$.

PROOF: For every innovation in a bin, the next state belongs to one of the intersected cells; its error lies in the corresponding stored interval and hence in their hull. Integration against the true bin probabilities preserves order. Backward

induction in the residual identity proves (23). Each displayed intersection combines two sets known to contain the same real value. Substituting $J^\pi = h + e$ in the action difference proves the final statement. An empty intersection indicates an invalid numerical calculation and must stop acceptance, rather than licensing a selected midpoint. *Q.E.D.*

This construction stores one-state bands rather than a table over state pairs. It sacrifices dependence between two trajectories when taking separate hulls; in particular, actor ambiguity and broad residual ranges can make it conservative. For locally Lipschitz critics, a centered residual evaluation can be intersected with the natural interval. A valid generalized-gradient enclosure suffices for the line-segment mean-value bound of a ReLU critic. This is not permission to differentiate a discontinuous acquired actor. On one cell its action is fixed; across cells the cache explicitly retains the alternative actions.

7.2. Signed sensitivity of the installed reference

The original investment family in Section 5 has a smooth zero-investment reference. Its stage and terminal state costs are continuously differentiable, including the squared shortage term. Let μ denote the zero-noise drift and γ the action exposure. For a state box X_0 , propagate boxes X_{j+1} containing $\mu(X_j, 0) + sZ$ for the complete innovation support. Starting from a terminal gradient enclosure $\mathcal{G}_r \supset \nabla g(X_r)$, put

$$\mathcal{G}_j \supset \nabla c(X_j, 0) + \beta D_x \mu(X_j, 0)^\top \mathcal{G}_{j+1}. \quad (24)$$

The inclusion means outward interval multiplication and addition. It is evaluated backward along forward state tubes, with the true sparse Jacobian.

PROPOSITION 10—Signed reference certificate: *The interval $\mathcal{G}_0$ encloses ∇J_0^0 throughout X_0 . For a nonnegative proposed action a , enclose the gradient of the remaining zero policy at every state $\mu(x, \theta a) + sZ$, $0 \leq \theta \leq 1$. If $[v_-, v_+]$ encloses the average projection $\mathbb{E}[\gamma^\top \nabla J_{t+1}^0]$ over that segment, then*

$$A_t^0(x, a) \in a^2(1 + 4a^2) + \beta a[v_-, v_+]. \quad (25)$$

The signed interval may be intersected with Proposition 9. The construction is valid for the complete zero reference; it does not differentiate a subsequently changed, cellwise constant policy.

PROOF: Differentiate the finite fixed-policy recursion. On the invariant compact domain, the derivatives are bounded, so differentiation and the finite conditional expectations may be interchanged. For every shock path the chain rule has the enclosed Jacobian form in (24); backward induction and expectation preserve the enclosure. The fundamental theorem of calculus along the first action segment gives

$$(P_t^a - P_t^0)J_{t+1}^0(x) = a \int_0^1 \mathbb{E}[\gamma^\top \nabla J_{t+1}^0(\mu(x, \theta a) + sZ)] d\theta.$$

The state-only current costs cancel and the remaining current difference is $a^2 + 4a^4$. This proves (25). Partitioning the first innovation into bins and integrating valid projected gradient hulls with their exact probabilities preserves the same argument. *Q.E.D.*

PROPOSITION 11—Reference stability and computational work: *In the dimension-normalized investment family, for every $d \geq 2$,*

$$\|D_x \mu\|_\infty \leq \kappa := 11/16, \quad \|\nabla c(\cdot, 0)\|_1 \leq 15/2, \quad \|\nabla g\|_1 \leq 10.$$

Consequently, with r stages remaining,

$$\text{Lip}_\infty(J_r^0) \leq \frac{15}{2} \sum_{j=0}^{r-1} (\beta\kappa)^j + 10(\beta\kappa)^r \leq \frac{1920}{91}, \quad \beta = 15/16. \quad (26)$$

For a partition with N leaves, A candidate and covering intervals per leaf, and q innovation bins, the literal cached range-query implementation uses

$$O(TAqN^2d)$$

cell-membership work, in addition to primitive and critic evaluation. Signed reference tubes over a full sweep use $O(ANqdT^2)$ work and can stream their innovation children. Their working state storage is $O(NdT)$ in addition to stored critics, cached bands and retained $O(TAN)$ endpoint records. No q^T recursion or full state-pair table is required.

PROOF: In row j , the only state derivatives of the drift are $1/2 + (1 - x_{j+1})/16$ and $1/8 - x_j/16$, both nonnegative. Their sum is at most $11/16$. The target, cyclic imbalance and shortage terms have gradient ℓ^1 bounds $5/2$, 1 and 4 for current cost; the terminal target bound is 5 . This gives the stated constants. The chain rule yields $L_r \leq 15/2 + \beta\kappa L_{r-1}$ with $L_0 \leq 10$. Its geometric expansion and $\beta\kappa = 165/256$ prove (26).

A queried next-state box is tested against N leaves in d coordinates. There are $O(TANq)$ such root-bin queries. The backward cache therefore has the displayed membership cost. Each gradient tube with r remaining dates visits $O(Nr)$ state rows and $O(Ndr)$ Jacobian coordinates for one first-innovation bin. Summing r across current dates gives $O(ANqdT^2)$. Streaming bins requires storage for one forward tube and its backward gradients, not all future shock combinations. *Q.E.D.*

The stability constant is a bound for this smooth reference, not a dimension-free complexity theorem for solving arbitrary dynamic programs. Resolving a desired policy or all-state accuracy may require the leaf count, network width and action cover to grow. The membership bound explicitly charges a quadratic leaf-count dependence. The executed locally split partition removes a tensor-product representation; it does not prove that a fixed 32-leaf partition resolves every eight-dimensional economic policy.

7.3. What a tighter interval establishes

For fixed candidates, let $C_t(C)$ be the greater of $L_t(C)$ and the minimum of their lower endpoints, including the incumbent's zero. Then

$$U - L = (U - C) + (C - L), \quad U - C \geq 0, \quad C - L \geq 0. \quad (27)$$

The first term measures uncertainty remaining between candidate lower information and the selected upper bound; the second measures the additional continuous-cover allowance. Their maxima need not add to the maximum of their sum. Actor ambiguity, acquisition width, innovation enclosures and arithmetic can enter both terms, so these are not separately estimated causal error components.

More detailed refinement can be recorded without such a causal interpretation. Keep the candidate actions fixed, replace each candidate upper bound by the minimum of its old and refined bounds, and replace the global lower bound by the maximum of its old and refined bounds. Then $U_r \leq U_{r-1}$, $L_r \geq L_{r-1}$, and the certificate improvement at step r is exactly

$$(U_{r-1} - L_{r-1}) - (U_r - L_r) = (U_{r-1} - U_r) + (L_r - L_{r-1}) \geq 0.$$

State subdivision, action-cover subdivision, innovation refinement and higher-precision arithmetic can each supply an independent valid refinement. The present execution records the cache-to-signed-tube refinement and both terms of (27); it does not label an unexecuted six-way refinement experiment as measured evidence.

8. FROM RETURNED POLICIES TO ECONOMIC RESOURCE DECISIONS

An all-state upper bound and a comparison of actual policy costs answer different economic questions. We now connect the policies returned by a construction frontier to direct cost inference and then to a choice of implementation resources. The construction data select the policies; independently generated evaluation paths assess them. This separation permits a valid comparison of first-crossing policies with different resolutions.

8.1. Simultaneous inference after construction

Let $\mathcal{H}$ denote all construction information: primitive inputs, numerical labels, fitted continuations, returned policies and complete certificate ladders. Conditional on $\mathcal{H}$, fix a finite catalogue of implemented-policy contrasts. Each implementation includes its observation rule, original witness or nearest-node selector, true-state feasible repair and action quantization. For a contrast i , let D_i be the difference in expected discounted costs under a specified initial law. A common-innovation path score evaluates the two policies on their own endogenous trajectories. It is not the difference between two regret bounds.

Suppose the initial and innovation uniforms are partitioned into equal bins. Independent uniform bin indices are drawn, and interval simulation encloses the score for every continuous realization inside the drawn bins. All unresolved state,

selector and repair branches remain in the enclosure. If $[A_i, B_i]$ is a deterministic support interval, clip the path endpoints to that support. Write $\underline{m}_i, \bar{m}_i$ for outward sample-mean bounds on the lower and upper endpoint samples and $v_{i,L}, v_{i,U}$ for upper bounds on their unbiased sample variances. For $n \geq 2$ define

$$R_i(v) = \sqrt{2v\ell/n} + \frac{7(B_i - A_i)\ell}{3(n-1)}, \quad \ell \geq \log(4K/\alpha), \quad (28)$$

where K bounds the number of sampled contrasts.

THEOREM 12—Construction-conditional economic intervals: *Under the stated independent-bin model, conditional on any valid construction history $\mathcal{H}$, with probability at least $1 - \alpha$ all catalogue contrasts satisfy*

$$D_i \in [l_i, u_i] := [\max\{A_i, \underline{m}_i - R_i(v_{i,L})\}, \min\{B_i, \bar{m}_i + R_i(v_{i,U})\}]. \quad (29)$$

The same coverage holds unconditionally. Subsequent selection among these covered contrasts, or evaluation at any deterministic implementation-price vector, needs no additional union bound. This conclusion applies only to policies and initial laws included in the catalogue; it does not create direct-cost observations for every unstudied tolerance or initial state.

PROOF: For each drawn bin sequence, conditional integration over its continuous realizations places the exact path expectation between the interval endpoints. Uniform mixing over the equal bins recovers the original continuous law. The lower endpoint expectation is thus no larger than D_i , and the upper endpoint expectation is no smaller. Apply the bounded empirical-Bernstein inequality of [Maurer and Pontil \(2009\)](#) to the endpoint samples and take the union bound, with outward arithmetic included in the mean, variance and radius. Conditional validity for every $\mathcal{H}$ implies unconditional validity by integration. On this one simultaneous event, all ensuing deterministic comparisons of the covered quantities hold together. *Q.E.D.*

In particular, selecting the first policy to meet a specified certificate is permitted when that selection uses construction records rather than the subsequent evaluation paths. Repeated construction clocks are not independent draws from a neural-optimizer population. The statistical randomness here belongs to the evaluation

design. A fixed pseudorandom stream implements a reproducible experiment; it is not itself a proof of the independent-bin assumption.

8.2. Replacement-fee identification

Write $D = J^W - J^F$. If witness is installed, replacing it by FVI changes total cost by $k - D$, where $k \geq 0$ is the replacement fee. If FVI is installed, the change is $k + D$. Thus the two directional gains before fees are D and $-D$.

COROLLARY 13—The break-even fee frontier: *On an event where $D \in [l, u]$, the two directional gains belong to $[l, u]$ and $[-u, -l]$. The unknown absolute cost difference satisfies*

$$|D| \in [\text{dist}(0, [l, u]), \max\{|l|, |u|\}]. \quad (30)$$

For every $k \geq \max\{0, u, -l\}$ neither replacement strictly lowers expected cost. By contrast, $k \geq \max\{0, l, -u\}$ only removes a certified strictly profitable replacement. The two thresholds generally differ. Strict inequalities in the first criterion certify that both replacements strictly increase cost.

PROOF: Subtract the gain intervals from k and compare the resulting interval endpoints with zero. Taking the image of $[l, u]$ under the absolute-value map gives (30). The assertions hold simultaneously for every nonnegative fee on the original coverage event. *Q.E.D.*

The fee frontier is more informative than a conclusion at one generous charge. The earlier 1/64 replacement decision is retained with its original interpretation and confidence account. The new analysis reports the two directional gain intervals and both thresholds instead of selecting a favorable fee after seeing the data. There is no welfare calibration hidden in the units: a decision maker must supply the economically relevant charge, or use the full frontier to see what remains unidentified.

8.3. Observation costs on the same bit grid

Let $\mathcal{B} = \{4, \dots, 16\}$ be the available bit precisions. For one fixed returned controller, write J_b for the expected cost of its actual b -bit, robustly repaired and quantized implementation. Use $b_0 = 16$ as a common reference and obtain simultaneous intervals for $D_b = J_b - J_{b_0}$. The identity interval at $b = b_0$ is exactly $[0, 0]$. Let $s_b = db \sum_{t < T} B_t$ be the discounted number of coordinate-bits purchased; the net cost is $J_b + \lambda s_b$.

THEOREM 14—Certified choice of actual net implementation cost: *Suppose all $D_b \in [l_b, u_b]$ simultaneously. For any $\lambda \geq 0$ define*

$$L(\lambda) = \min_b \{l_b + \lambda s_b\}, \quad U(\lambda) = \min_b \{u_b + \lambda s_b\}, \quad (31)$$

$$\hat{b}(\lambda) = \min \arg \min_b \{u_b + \lambda s_b\}.$$

Then, simultaneously for all $\lambda \geq 0$,

$$0 \leq J_{\hat{b}} + \lambda s_{\hat{b}} - \min_{b \in \mathcal{B}} (J_b + \lambda s_b) \leq U(\lambda) - L(\lambda). \quad (32)$$

The lower and upper functions are finite piecewise-affine envelopes. All changes in their active lines occur at nonnegative intersections of catalogue lines, with exact comparisons deciding ties. The smallest selected precision $\hat{b}(\lambda)$ is weakly decreasing in λ .

PROOF: Subtract the common J_{b_0} . The selected implementation has net difference at most $U(\lambda)$, and the best catalogue implementation has net difference at least $L(\lambda)$. This proves the regret bound pointwise on the one simultaneous event and hence for every λ . A minimum of finitely many affine functions changes its active affine expression only at an intersection. For $\lambda_2 > \lambda_1$, sum the two minimizing inequalities to obtain $(\lambda_2 - \lambda_1)(s_{\hat{b}(\lambda_2)} - s_{\hat{b}(\lambda_1)}) \leq 0$. Since s_b increases strictly with b , the precision comparative static follows. *Q.E.D.*

This result makes a statement about *actual* net cost within the declared bit catalogue. It differs from minimizing a uniform all-state loss bound. Both criteria are useful: the latter gives distribution-free protection relative to the full-information

optimum, while (32) uses a specified initial law to compare feasible implementations of one returned policy. Neither criterion gives an optimum over arbitrary sensor technologies or all policies.

For a fair comparison the conventional actor receives its own acquisition allowance. Suppose its nearest-node policy has a certified selected-policy account at a reported state, obtained from the graph modulus L_t^{gr} , and its actual interpolated critic has modulus L_t^f . Transporting the same selected node action to the true state, then using cell-wide capacity repair, adds at most

$$(L_t^{\text{gr}} + L_t^f + 2D_t K_t^A) r(b) + D_t \Delta_t. \quad (33)$$

Indeed, transport the feasible node pair directly to the true state, bound its node distance by the reported-state distance plus $r(b)$, and bound $f_t(\hat{x}) - f_t(x)$ by $L_t^f r(b)$. Robust repair adds $D_t(2K_t^A r(b) + \Delta_t)$. This proves (33) without assuming actor continuity. For a witness critic $L_t^f = L_t^{\text{gr}}$, it reduces to the preceding witness allowance. The empirical comparison therefore evaluates the all-state-account minimizer, the interval upper-net-cost selector, and a descriptive midpoint selector on the *same* integer bit and price grid. The midpoint selector is not called the true optimum.

9. PROSPECTIVE ECONOMIC STOPPING AND LEARNED STRUCTURE

A policy certificate determines whether a proposed change is safe at every state. An economic stopping rule asks when the returned complete policy has attained a specified expected-cost target under the initial law. Neither question is answered by choosing the best recorded policy after all computation has finished.

9.1. A first-attainment service

Fix an installed policy π^0 , a target $q \in (0, 1)$, a finite resource ladder, and cumulative sample sizes at each stage. At a stage, construct a critic from primitives, transfer the old policy exactly to any refined partition, and execute a verified simultaneous sweep. Freeze the resulting complete policy before drawing that stage's fresh evaluation stream. At each declared look form a confidence interval for

$$D_{k,q} = \mathbb{E} J_0^{\pi^k} - q \mathbb{E} J_0^{\pi^0}. \quad (34)$$

Return the policy at the first nonpositive upper endpoint. A strictly positive lower endpoint ends the current stage and advances to the next resource stage. An unresolved interval leads to the next declared look. Exhausting the last stage returns the safe incumbent and an explicit budget-exhaustion status. It is not a conclusion that the target is unattainable.

THEOREM 15—Validity at prospective return: *Suppose each accepted sweep satisfies Theorem 8. At each declared stage and look, assume the interval for (34) has conditional failure probability at most $\alpha_{k,l}$ given the information preceding the stage's fresh evaluation stream. If $\sum_{k,l} \alpha_{k,l} \leq \alpha$, then, with probability at least $1 - \alpha$, every successful return satisfies its stated expected-cost target. Every returned incumbent, including one returned after budget exhaustion, is statewise no worse than the original installed policy. No independence between different methods or different looks is required for the union bound. Each within-stage endpoint sample must satisfy its stated conditional sampling assumptions.*

PROOF: Condition on the history preceding each executed stage. Its frozen policy and fresh independent-bin stream satisfy the fixed-policy interval guarantee. Integrating conditional failure probabilities and taking the union over the prespecified potential looks gives a simultaneous event of probability at least $1 - \alpha$. The first-crossing rule selects one already covered interval on this event, so its nonpositive upper endpoint implies (34) is nonpositive. Unexecuted later looks need not be computed. Statewise safety follows by composition of the deterministic sweep inequalities and exact policy transfer, independently of the economic inference event. *Q.E.D.*

The theorem also permits a countable predetermined ladder with summable failure allowances. The experiment uses the finite ladder specified below, not an unimplemented anytime extension. A process clock begins before construction and ends after its durable output. Failed stages, own-policy evaluation and inference are charged to that service. A method reaching a target early must not be charged for hypothetical later stages, and a method failing early must not have its completed work omitted.

For the implemented bounded-sample account, each path is enclosed over its complete forty-bit initial and innovation cells. The last innovation is integrated analytically. The interval in Theorem 12 applies at each declared look, using the known support and outward sample moments. The new family limit is 2,048 intervals with logarithm upper bound 14 and error 1/100; the planned maximum is 1,188 intervals. Earlier looks reuse prefixes within a stage and are covered by the same finite union. Distinct stages use fresh streams. A fixed pseudorandom seed establishes reproducibility under the declared independent-bin model, not mathematical independence of a deterministic sequence.

COROLLARY 16—Economic adoption at return: *Suppose the returned policy has cumulative gain $G = \mathbb{E}J_0^{\pi^0} - \mathbb{E}J_0^\pi \in [l, u]$ on the simultaneous event. For every replacement charge $\tau \geq 0$, computation price $\lambda \geq 0$, and recorded complete work $W \geq 0$, its net gain belongs to*

$$[l - \tau - \lambda W, u - \tau - \lambda W].$$

Thus $\tau + \lambda W < l$ certifies profitable adoption and $\tau + \lambda W > u$ rules out profitable adoption. These comparisons hold simultaneously over the entire nonnegative price plane on the original event.

PROOF: Subtract the same known quantity $\tau + \lambda W$ from both endpoints of the gain enclosure. The assertion is pathwise on the simultaneous event, so choosing prices after observing recorded work requires no extra confidence allocation. *Q.E.D.*

This is a decision at return, not a claim that the executed relative-cost stopping rule optimizes all possible prices or future computation schedules. The units of λ must convert recorded work into the model's cost units. In an uncalibrated theoretical economy the price plane is an explicit economic sensitivity analysis, not an estimated welfare magnitude.

9.2. Learning an approximately null direction

Suppose the transition is $\mu_t(x, 0) + Ba + sZ$ with unknown $B \in [0, 1]^d$, known nonlinear drift and known bounded additive action-invariant innovation law. Fit

a direction $\hat{v}$ on training observations. Independent validation observations Y_i estimate B coordinatewise; each coordinate has mean B_j and range length at most w . Let the recorded mean enclosure be $[\underline{m}_j, \overline{m}_j]$ and let δ_{bin} cover numerical bin displacement. Form

$$\mathcal{K}_N = \prod_{j=1}^d \left[\max\{0, \underline{m}_j - r_N\}, \min\{1, \overline{m}_j + r_N\} \right], \quad r_N = w \sqrt{\ell / (2N)} + \delta_{\text{bin}}. \quad (35)$$

Only validation data, the fitted direction and the declared parameter prior enter the gate.

THEOREM 17—Confidence-qualified action-null correction: *For a finite collection of H validation boxes, each with at most $d_{\max}$ coordinates, $2Hd_{\max}e^{-\ell} \leq \alpha_B$ gives simultaneous coverage of their true exposures with probability at least $1 - \alpha_B$. On this event, for every fitted direction, amplitude M , and nonnegative feasible action pair,*

$$|(P_B^a - P_B^b)M(\hat{v}^\top y + c)_+| \leq |M| \sup_{B' \in \mathcal{K}_N} |\hat{v}^\top B'| |a - b|. \quad (36)$$

A whole-cell critic-advantage upper bound evaluated uniformly over the same box, plus the discounted right-hand side of (36), is a valid upper bound for the corresponding true advantage with the nuisance removed. Accepting only when that bound is nonpositive preserves safety simultaneously for all tested amplitudes and actions. The simplification is rejected whenever its uncertainty allowance prevents the gate from passing.

PROOF: The bounded-coordinate inequality of [Hoeffding \(1963\)](#) and a union bound give coverage of (35); numerical mean and bin allowances preserve containment. On that event, use a common innovation to couple the two actual kernels. Their next-state difference is $B(a - b)$. The positive-part map is one-Lipschitz, so its projected difference is bounded by $|\hat{v}^\top B| |a - b|$ for every realization. Integration proves (36). The box event controls all linear projections, not merely a preselected direction. Uniformly enclosing the rest of the critic advantage over that same parameter set prevents an unobserved true-parameter oracle from entering the remaining calculation. Subtracting a nuisance difference cannot increase the upper bound by more than its absolute allowance. *Q.E.D.*

The experiment uses $w = 1/2$, $\ell = 14$, 27 validation boxes and at most eight coordinates, giving error at most $1/200$. The direction is fitted on 256 observations and then quantized; the exact stored direction is used in (36). Validation sizes are 512, 4,096 and 32,768. The guarantee concerns an unknown action-exposure vector in a maintained structural model. It does not infer an arbitrary unknown transition law, and coverage failures must remain in the evidence rather than being discarded by reference to the simulator’s known truth.

An empty validation box invalidates the parameter qualification and rejects the simplified certificate. It cannot authorize a policy change.

10. PROSPECTIVE RETURNS IN THE NONLINEAR INVESTMENT ECONOMY

10.1. Design

Tasks are $(d, T) = (2, 2), (4, 4), (8, 6)$. Each method/task/target combination is a separate from-primitives service. Targets $q = 9/10$ and $49/50$ require ten- and two-percent expected-cost reductions relative to installed zero investment. The stopping statistic is the signed contrast $J(\pi) - qJ(0)$, not a ratio of plug-in mean estimates. Two resource stages use respectively 256 and 1,024 training states, 32 and 128 non-tensor leaves, and four and eight innovation bins. Actions are downward quantized at $1/4096$ using the lower capacity over the complete leaf. Partition geometry is chosen before the generator and is nested between stages.

Trained ReLU, quadratic and ExtraTrees fitted-value methods use their own future targets. The neural architecture and training are described in Section 6. ExtraTrees uses 16 trees, depth at most eight and minimum leaf size three; quadratic FVI fits a ridge-regularized quadratic. The original two-dimensional frontends are reconstructed at $N = 8, 16$, $K = 4$ and $M = 4$. Every method receives nine common robust candidate actions and one fitted proposal, as well as eight intervals covering the full continuous feasible action set. The new algebraic neural action solver is evaluated separately on saved critics; it was not substituted for the frozen experimental construction menu.

After one complete simultaneous sweep at a stage, inference uses cumulative looks of 4,096, 16,384 and 65,536 interval paths. It stops on the first covered target crossing, advances a stage at a strictly positive lower target contrast or exhausted

look budget, and never executes a later stage after success. All computation, including failed stages and each method’s own inference, is retained. Three isolated sequential processes repeat identical training and inference seeds, with rotated method order, a fixed available CPU core and one numerical-library thread. These measure reproducibility and clock variation, not a population of independently trained networks. Clock frequency is recorded where available but not controlled.

10.2. *The refinement and the complete returned policies*

The first cohort uses the residual/value cache. Its complete 66-process record contains 39 target attainments and 27 budget exhaustions. In the eight-dimensional task it makes no policy change. The separate sensitivity extension was motivated by that completed result and frozen before its own production and fresh inference. It retains the same targets, fitted constructions and common actions, while intersecting the old cache comparison with the signed reference interval of Proposition 10. This chronology matters: the extension is not described as prespecified before observing the primary cohort.

TABLE I

OUTCOMES BEFORE THE SIGNED REFERENCE REFINEMENT

d	T	q	Stages	Paths	Status	Last-pass changes
2	2	49/50	1	16384	Hit	27
2	2	9/10	2	65536	Hit	93
4	4	49/50	2	16384	Hit	90
4	4	9/10	2	16384	Exhausted	90
8	6	49/50	2	65536	Exhausted	0
8	6	9/10	2	4096	Exhausted	0

This primary-cache cohort is not dropped after its limitations become visible. These are the neural rows; the candidate generators return policy-identical final arrays within each primary task and target.

Exhausted means the declared finite service did not certify its target, not that improvement is impossible. All own construction, inference and failed-stage work remains charged.

All 66 signed-reference services attain their targets at the first resource stage. This means one complete all-date sweep on 32 leaves, not T exact sweeps and not

TABLE II

ACTUAL COSTS AT THE PROSPECTIVE NEURAL-POLICY RETURN

d	T	q	Paths	Expected cost	Installed minus returned
2	2	49/50	4096	[0.5338, 0.6834]	[0.0622, 0.2408]
2	2	9/10	16384	[0.5950, 0.6503]	[0.1294, 0.1777]
4	4	49/50	4096	[0.8975, 1.0658]	[0.1199, 0.3796]
4	4	9/10	4096	[0.8974, 1.0658]	[0.1177, 0.3775]
8	6	49/50	4096	[1.3427, 1.5356]	[0.0918, 0.4154]
8	6	9/10	16384	[1.4216, 1.4788]	[0.2145, 0.2978]

N is the trained ReLU generator. Every row is a separately executed target service with the signed reference certificate. Endpoints are rounded outward. The interval covers the complete implemented policy under the original continuous law. Timing repetitions have identical policy and inference identities and are not additional economic observations. Other generators and every earlier-stage outcome are retained in the supplement.

attainment of the Bellman optimum. For the neural eight-dimensional, six-date controller, the changed leaf counts by date are $(26, 24, 24, 20, 14, 0)$. Its identified cost reduction is generated by nonterminal decisions; it is not a terminal-repair experiment relabeled as a long-horizon result. At the ten-percent target its cumulative gain interval is $[0.21452658, 0.29774911]$, with endpoints rounded outward. The complete expected cost interval is $[1.42162445, 1.47873990]$.

By Corollary 16, the eight-dimensional return supports every replacement-and-computation charge below its lower gain endpoint, in the model's normalized units. A charge above its upper gain endpoint rules out profitable adoption. The interval between them remains unresolved. This translates the returned policy into a full economic price sensitivity, rather than choosing a favorable fee after inspecting one estimate.

The directed comparison, all-state optimality account and initial-distribution cost interval continue to answer different questions. At date zero of the eight-dimensional neural service, the maximum candidate interval width falls from more than 14 to less than 0.610 after the signed refinement. The full-action greedy allowance remains about 1.021, and the inherited finite-sweep all-state bound is much looser than the observed expected cost gain. The experiment does not

TABLE III

WHERE THE SIGNED REFERENCE CERTIFICATE CHANGES DECISIONS

Date	Changed	Base width	Refined width	$U - C$	$C - L$	$U - L$
0	26	14.094	0.610	0.604	0.418	1.021
1	24	12.158	0.594	0.588	0.407	0.994
2	24	10.094	0.569	0.562	0.388	0.950
3	20	7.892	0.530	0.521	0.358	0.879
4	14	5.543	0.466	0.453	0.310	0.762
5	0	1.055	1.055	0.640	0.074	0.713

Eight-dimensional state, six dates, trained ReLU, first stage, 32 non-tensor leaves. Widths are maxima over candidate enclosures; decomposition entries are maxima over leaves, rounded upward. The last row uses analytic terminal differences rather than differentiating an acquired actor. A tighter comparison certificate is not itself a new policy-cost observation.

claim that 32 leaves or one sweep solve the continuous-action Bellman problem accurately at every state. It does establish the declared prospective expected-cost improvement for the complete implemented policy, together with deterministic statewise incumbent protection.

10.3. Method comparisons and complete work

The trained ReLU proposal changes no deployed cell beyond the nine common actions in either cohort. The only tenth-proposal effect is one compiled-witness cell in the two-dimensional sensitivity task, repeated across targets and timing runs. These identities are checked explicitly rather than inferred from similar cost intervals.

The conventional generators receive the same acquired states, common candidate actions, signed reference information and economic stopping rule. The audit reconstructs the trained proposal and then removes the tenth candidate from the recorded gate to determine whether that proposal changes the selected action. This is a deterministic candidate-attribution diagnostic, not a fresh policy-cost experiment. Exact deployed-policy identities are checked across generators and across timing repetitions.

TABLE IV

COMPLETE PROCESS WORK TO THE TEN-PERCENT COST TARGET

d	T	Method	Min (s)	Median (s)	Max (s)	Output KiB	RSS MiB
2	2	W	0.781	0.808	1.345	739	42.0
2	2	E	1.678	1.737	1.740	1100	110.0
2	2	Q	0.695	0.701	0.745	781	42.3
2	2	N	1.389	1.409	1.431	788	50.6
2	2	F	0.807	0.809	0.853	731	42.0
4	4	E	11.515	11.684	11.781	1286	107.3
4	4	Q	5.576	5.582	5.583	489	39.0
4	4	N	9.073	9.087	9.238	512	50.7
8	6	E	31.833	31.886	35.303	2588	112.3
8	6	Q	21.027	21.243	21.882	1350	43.8
8	6	N	26.714	26.771	30.501	1400	51.8

N: trained ReLU; Q: quadratic FVI; E: ExtraTrees FVI; W: compiled witness; F: tensor FVI. Three isolated sequential processes use identical seeds within a method. Each clock includes its own imports, construction, verification, inference and durable output. Compare methods within a task runner, not across runners. Affinity and numerical-library threads are fixed; clock frequency is not controlled. Output excludes prior-stage archives outside the service; RSS is process peak memory. Both targets and all repetitions appear in the supplement.

In the four- and eight-dimensional sensitivity tasks, the generators return policy-identical arrays within each target. In the two-dimensional task the compiled-witness return differs slightly, while the trained ReLU, quadratic, ExtraTrees and tensor-FVI returns coincide. Thus the policy improvement is not evidence that the neural generator dominates these alternatives. At the ten-percent target, quadratic FVI has the lowest recorded median complete-process work in each task. The neural backend supplies a genuinely trained nonlinear continuation and analytic integration, but those facilities must not be confused with an identified economic ranking.

The clocks include imports, primitive construction, critic fitting, acquired-policy preparation, verification, own inference, serialization and durable outputs. Method

comparisons are within each task’s controlled runner; clocks from different task runners are not a hardware-controlled scaling regression.

10.4. *Neural action witnesses and learned null directions*

TABLE V

EXACT LATTICE SEARCH ON SAVED FITTED NEURAL CRITICS

d	T	Full lattice	Candidates	Old index	Exact index	Critic decrease
2	2	896	10	279	303	0.0000800
4	4	768	4	191	169	0.0000289
8	6	640	6	559	555	0.0000012

One declared cell center and its saved next-date critic per task. Rational isolation and comparison agree with exhaustive rational lattice minimization. The reported decrease is in that fixed approximate Bellman objective, not actual economic policy cost. These solver regressions neither retrain a critic nor produce new policy-cost observations.

The algebraic action solver gives exact fitted-objective minimizers on the full deployed lattice in its retained regressions, using only a small subset of lattice indices. This is a constructive neural-search result. Its objective decreases concern the saved approximate continuation at a specified cell center, not the expected cost of a new deployed controller. All historical inference remains attached to the policies that were actually evaluated.

The sample-split study fits nine directions across dimensions two, four and eight and three independent direction-fitting seeds. Three validation sizes give 27 confidence boxes; three amplitudes on 64 acquired leaves give 81 cases and 5,184 cell-case decisions. All 27 recorded boxes contain the simulator exposure. The corrected gate makes 165 safe changes. No strictly harmful plug-in decision is identified in this learned-direction catalogue. That zero count is retained; it is not replaced by the large harmful count from the earlier, different known-perturbation experiment.

The primary policy-cost, sensitivity policy-cost and learned-exposure families have errors 1/100, 1/100 and 1/200, respectively. Their joint union error is at

TABLE VI

VALIDATION-QUALIFIED LEARNED DIRECTIONS

d	Validation rows	Projection-bound range	Corrected changes	Harmful plug-in
2	512	[0.0660, 0.0670]	36	0
2	4096	[0.0241, 0.0271]	54	0
2	32768	[0.0121, 0.0141]	75	0
4	512	[0.0608, 0.0636]	0	0
4	4096	[0.0217, 0.0232]	0	0
4	32768	[0.0081, 0.0102]	0	0
8	512	[0.0586, 0.0622]	0	0
8	4096	[0.0207, 0.0236]	0	0
8	32768	[0.0078, 0.0101]	0	0

Each row aggregates three independently fitted directions and three amplitudes on 64 leaves per case.

The projection range reports the minimum and maximum robust upper allowance across fitted directions, not a confidence interval for a population mean. All 27 exposure boxes cover the simulator exposure in this record. No strictly harmful plug-in choice is identified here; the earlier known-perturbation catalogue is a different diagnostic and remains unchanged.

most 1/40. This addition combines covered statements; it does not make identical policies, timing repetitions, amplitudes or reused prefixes independent observations. The learned-direction conclusion remains conditional on the known drift and maintained additive action-invariant innovation structure.

10.5. *Relation to the original evidence*

The earlier witness, FVI, acquisition, common-repair, exact-null and adaptive-grid results remain in the complete development article and its supplement. In particular, earlier adverse witness-cost comparisons are not erased by the improvement of a newly installed zero controller. The new first-attainment services replace retrospective catalogue work as the primary work-to-target endpoint for this experiment; the historical catalogue remains a valid account of a different executed procedure. The full source and result archive preserves both and states which economic policy and law each interval actually covers.

11. CONCLUSION

Neural Bellman Operators connect an own-future continuation to a feasible action and an economic cost account. The constructive backend retains the original Bellman-accuracy objective. The trained backend now has analytic innovation integration and a terminating exact action-lattice search, while its proposals remain subject to whole-cell true-advantage verification. These are complementary parts of the same numerical method, not interchangeable claims about the name of a function class.

The non-tensor cache and signed reference calculation turn the policy account into an executable prospective service. Their proofs retain the cost of leaf membership, critic evaluation, state dimension and horizon. The declared expected-cost targets are attained in the sensitivity experiment, with genuine nonterminal action changes and complete own-service clocks. Conventional generators share the same verification treatment, and policy identities make their role observable. The finite experiments identify successful returns and their costs; they do not establish universal neural dominance or calibrated welfare effects.

The full-action gap, the actual expected cost and the price of acquiring or verifying a policy remain separate economic quantities. Keeping all three explicit permits a decision maker to choose the appropriate contract without treating fitting accuracy, certificate sharpness or a favorable timing observation as a substitute for the cost of the implemented policy.

APPENDIX A: DEVELOPMENT AND REPRODUCIBILITY MAP

The complete development article and proof companion preserve every preceding theorem, economic application and numerical comparison, including adverse results. The prior active article and supplement are also retained as separate development documents. The present main article reorganizes the original constructive core around trained action witnesses, non-tensor verification and prospective economic stopping; it does not replace the NBO subject with a different model.

The publication audit regenerates all unique recorded certificates from their saved fitted critics, replays all stopping decisions and verifies original moment

bounds against exact rational sample moments. New solver regressions compare rational action witnesses with exhaustive lattice minimization. These operations do not create new policy-cost samples or replace historical service clocks. The source freezes, ordinary manuscript sources, complete service registry, raw arrays, response and release manifests are included in the revision directory. Mathematical peer review remains distinct from executable regression and document reproduction.

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